

Project Introduction

Project Name

Restaurant Management System

Purpose

- The purpose of the Restaurant Management System is to help restaurants manage their daily operations in a structured and efficient way. The system will allow restaurant staff to manage menus, tables, customer orders, kitchen preparation, billing, payments, inventory, and reports from a centralized application.
- The system is intended to reduce manual work, avoid order and billing mistakes, improve communication between waiters, kitchen staff, cashiers, and managers, and provide better visibility into restaurant performance.
- **The first version** of the system will focus on a **single restaurant**. However, it will be designed in a way that it can later be extended into a SaaS-based platform where multiple restaurants can manage their own operations independently.

Problem Statement

Many restaurants manage daily operations using manual processes or disconnected tools such as paper notes, spreadsheets, phone calls, or verbal communication. This can create operational issues, especially during busy hours when multiple staff members need to coordinate quickly and accurately.

The major problems in the existing process are:

- Orders may be written incorrectly or missed.
- Kitchen staff may receive delayed or unclear order details.
- Waiters may not know the real-time preparation status of food items.
- Billing mistakes can occur if order items are not tracked properly.
- Table availability may not be updated in real time.
- Inventory may not be tracked properly, causing stock shortages or wastage.
- Managers may not get clear reports about sales, orders, payments, and stock usage.
- Communication between waiters, kitchen staff, cashiers, and managers can become difficult during peak hours.

The Restaurant Management System is intended to solve these problems by providing a centralized application to manage orders, kitchen preparation, billing, payments, inventory, table status, and reports in a structured and trackable way.

Target Users

- The target users of the Restaurant Management System are restaurant staff and management users who are involved in daily restaurant operations. These users include Restaurant Admin, Manager, Waiter, Chef/Kitchen Staff, Cashier, and Inventory Manager.
- Each user will access the system based on their role and assigned permissions. For example, waiters will use the system to create and manage customer orders, chefs will use it to view kitchen orders and update preparation status, cashiers will use it for billing and payments, and managers/admins will use it to manage operations, users, menu, reports, and restaurant settings.
- In future versions, the system may also support customers and delivery staff if online ordering and delivery features are added.

Business Goal

- The business goal of the Restaurant Management System is to improve the efficiency, accuracy, and visibility of restaurant operations by digitizing key activities such as order management, kitchen coordination, billing, payments, inventory tracking, and reporting.
- The system aims to reduce manual work, minimize order and billing errors, improve staff coordination, speed up service, and help restaurant management make better decisions using accurate operational and sales data.
- In the long term, the system should be designed so it can be extended into a SaaS-based platform where multiple restaurants can independently manage their own operations.

Project Scope

The scope of the Restaurant Management System is to define the features, modules, and operational areas that will be covered by the system. The first version will focus on managing the core operations of a **single restaurant**, including staff access, menu setup, table handling, dine-in order processing, kitchen tracking, billing, inventory, and basic reporting.

The system will be designed with future extensibility in mind, so that it can later support SaaS-based usage, multiple restaurants, and multiple branches if required.

In Scope

The first version of the Restaurant Management System will focus on managing the core operations of a single restaurant. The following features and modules will be included:

- User authentication and role-based access for Admin, Manager, Waiter, Chef/Kitchen Staff, Cashier, and Inventory Manager.
- Basic restaurant setup such as restaurant profile, contact details, tax settings, and service charge settings.
- Table management, including table creation, seating capacity, and table status tracking.
- Menu management, including menu categories, menu items, prices, descriptions, and item availability.
- Dine-in order management, including creating orders for tables, adding items, updating quantities, adding item notes, and sending orders to the kitchen.
- Kitchen order management, including viewing kitchen orders and updating item preparation status such as Pending, Preparing, and Ready.
- Billing and payment management, including bill generation, tax/service charge calculation, discount handling, payment recording, and receipt printing/downloading.
- Basic inventory management, including ingredients, stock entries, suppliers, reorder levels, low-stock tracking, and wastage recording.
- Basic reporting, including daily sales, order summary, payment summary, item-wise sales, and inventory reports.
- Basic audit/activity tracking for important actions such as order creation, item status updates, billing, payment, and cancellation.

Out Of Scope

The following features will not be included in the first version of the Restaurant Management System:

- Customer-facing online ordering website or mobile application.
- Customer registration, login, and order history.
- Delivery order management and live delivery tracking.
- Third-party food delivery platform integration such as Swiggy, Zomato, Uber Eats, etc.
- Online payment gateway integration.
- Loyalty points, coupons, gift cards, and membership programs.
- Full SaaS onboarding for multiple restaurants.
- Multi-branch management.
- Advanced accounting and payroll management.
- Advanced recipe-based automatic stock deduction.
- Advanced purchase order and vendor approval workflow.
- Advanced analytics dashboard with forecasting or AI-based recommendations.
- Offline mode for order taking and billing.
- Integration with external hardware such as barcode scanners, kitchen display systems, cash drawers, or POS terminals.
- Advanced reservation and table booking system.

Future Scope

The following features and enhancements may be considered in future versions of the Restaurant Management System:

- SaaS-based multi-restaurant support, where multiple restaurants can register and manage their own users, menu, tables, orders, billing, inventory, and reports independently.
- Multi-branch management, allowing a restaurant to manage multiple branches with branch-wise users, menus, inventory, orders, and reports.
- Customer-facing online ordering through a website or mobile application.
- Support for additional order types such as takeaway, delivery, and online orders.

- Delivery management, including delivery staff assignment, delivery status tracking, and customer address management.
- Online payment gateway integration for UPI, cards, wallets, or net banking.
- Customer loyalty features such as coupons, loyalty points, memberships, offers, and gift cards.
- Advanced inventory and purchase management, including recipe-based stock deduction, purchase orders, vendor approval workflows, expiry tracking, and ingredient cost calculation.
- Advanced reporting and analytics, including profit/loss reports, sales trends, peak-hour analysis, item performance, inventory consumption, and forecasting.
- External integrations with food delivery platforms, POS terminals, kitchen display systems, barcode scanners, notification services, and accounting software.
- Offline mode for order taking and billing with later data synchronization.
- Advanced table reservation and booking management.

User Roles and Permissions

Overview

The Restaurant Management System will use role-based access control to ensure that each user can access only the features and data required for their job responsibility. Each staff member will be assigned a role such as Admin, Manager, Waiter, Chef, Cashier, or Inventory Manager. Based on the assigned role and permissions, the system will allow or restrict actions such as managing menu items, creating orders, updating kitchen status, generating bills, accepting payments, and managing inventory.

User Roles

1. Admin
2. Manager
3. Waiter
4. Chef / Kitchen Staff
5. Cashier
6. Inventory Manager

Role Responsibilities

Admin

The Admin is responsible for managing the main setup and configuration of the restaurant system. This role can manage users, roles, permissions, restaurant settings, menu data, tables, taxes, discounts, and reports.

- Manage staff users
- Assign roles and permissions
- Manage restaurant settings
- Manage menu categories and menu items
- Manage tables
- Manage taxes and service charges
- View reports

Manager

The Manager is responsible for monitoring and controlling daily restaurant operations. This role can manage live orders, table status, staff coordination, order cancellations, discounts, and daily operational reports.

- Monitor active orders
- Monitor table status
- Assign or assist waiters
- Approve order/item cancellation
- Approve discounts if permitted
- Handle customer issues
- View daily reports

Waiter

The Waiter is responsible for taking customer orders and managing table service. This role can create dine-in orders, add items, add special instructions, send orders to the kitchen, view order status, and request bill generation.

- View assigned/open tables

- Create dine-in orders
- Add items to an order
- Add item notes or instructions
- Send order to kitchen
- View kitchen/order status
- Mark items as served
- Request bill

Chef / Kitchen Staff

The Chef or Kitchen Staff is responsible for preparing food items based on orders received from waiters. This role can view kitchen orders, view item notes, and update item preparation status.

- View kitchen orders
- View order item details
- View item preparation notes
- Mark item as Preparing
- Mark item as Ready

Cashier

The Cashier is responsible for billing and payment-related activities. This role can view billable orders, generate bills, apply allowed discounts, record payments, print receipts, and close paid orders.

- View billable orders
- Generate final bill
- Verify order items before billing
- Apply tax and service charge
- Apply allowed discount
- Record payment
- Print/download receipt
- Mark order as paid/closed

Inventory Manager

The Inventory Manager is responsible for managing restaurant stock, ingredients, suppliers, purchase entries, reorder levels, and wastage records.

- Manage ingredients/raw materials
- Add stock purchase entries
- Update stock quantity
- Manage suppliers
- Set reorder levels
- View low-stock items
- Record wastage
- View inventory reports

Permission Matrix

Feature / Action	Admin	Manager	Waiter	Chef	Cashier	Inventory Manager
Manage users	Yes	No/Limited	No	No	No	No
Manage menu	Yes	Limited	No	No	No	No
Manage tables	Yes	Yes	View	No	View	No
Create order	Yes	Yes	Yes	No	No	No
Send order to kitchen	Yes	Yes	Yes	No	No	No
Update kitchen status	No	Yes	No	Yes	No	No
Mark item as served	No	Yes	Yes	No	No	No
Generate bill	Yes	Yes	No	No	Yes	No
Accept payment	Yes	Limited	No	No	Yes	No
Manage inventory	Yes	No	No	No	No	Yes
View reports	Yes	Yes	No	No	Limited	Inventory only

Functional Requirements

Functional requirements define what the Restaurant Management System should do. These requirements are written module-wise so that the system can be understood, designed, developed, and tested clearly.

Authentication and User Management

This module will handle user login, staff user management, role assignment, and access control. The system should ensure that only authorized users can access the features allowed for their role.

- The system shall allow authorized staff users to log in using valid credentials.
- The system shall prevent users from accessing the system with invalid credentials.
- The system shall allow logged-in users to log out from the system.
- The system shall restrict access to features based on the user's assigned role and permissions.
- The system shall allow Admin to create staff users.
- The system shall allow Admin to update staff user details.
- The system shall allow Admin to activate or deactivate staff users.
- The system shall allow Admin to assign a role to each staff user.
- The system shall allow Admin to view the list of staff users.
- The system shall allow Admin to reset a staff user's password.
- The system shall allow users to change their own password.
- The system shall store user passwords securely using hashing.
- The system shall not allow deactivated users to log in.
- The system shall allow users to view their basic profile information.
- The system shall validate required user details such as name, email or mobile number, role, and status.
- The system shall prevent duplicate email or mobile number for staff users.
- The system shall record the created date and updated date for each staff user.

Restaurant Setup

This module will allow the Admin to manage the basic configuration and profile information of the restaurant. The restaurant setup information will be used across the system for billing, reports, and general restaurant identification.

- The system shall allow Admin to create or update the restaurant profile.
- The system shall allow Admin to manage restaurant details such as restaurant name, address, contact number, and email.
- The system shall allow Admin to upload or update the restaurant logo.
- The system shall allow Admin to configure tax settings.
- The system shall allow Admin to configure service charge settings.
- The system shall allow Admin to define restaurant opening and closing time.
- The system shall allow Admin to set the default currency.
- The system shall allow Admin to manage basic billing information that appears on customer receipts.
- The system shall allow Admin to activate or deactivate tax and service charge settings.
- The system shall use restaurant setup details while generating bills and reports.

Table Management

This module will allow Admin and Manager to manage restaurant tables and track table availability during restaurant operations. Table information will be used while creating dine-in orders and managing customer seating.

- The system shall allow Admin to create tables.
- The system shall allow Admin to update table details.
- The system shall allow Admin to delete or deactivate tables.
- The system shall allow Admin to define table number or table name.
- The system shall allow Admin to define seating capacity for each table.
- The system shall allow Admin or Manager to view the list of tables.
- The system shall display the current status of each table.

- The system shall support table statuses such as Available, Occupied, Reserved, and Cleaning.
- The system shall allow Waiter or Manager to select an available table while creating a dine-in order.
- The system shall automatically mark a table as Occupied when an active order is created for that table.
- The system shall allow Manager or Waiter to update table status when required.
- The system shall automatically mark a table as Available after the order is completed and payment is done.
- The system shall prevent creating multiple active dine-in orders for the same table at the same time.

Menu Management

This module will allow Admin and authorized Manager users to manage menu categories and menu items. Menu data will be used by waiters while creating orders and by the billing module while calculating order amounts.

- The system shall allow Admin to create menu categories.
- The system shall allow Admin to update menu categories.
- The system shall allow Admin to delete or deactivate menu categories.
- The system shall allow Admin to create menu items.
- The system shall allow Admin to update menu item details.
- The system shall allow Admin to delete or deactivate menu items.
- The system shall allow Admin to assign each menu item to a menu category.
- The system shall allow Admin to define menu item name, description, price, and availability.
- The system shall allow Admin to upload or update a menu item image if required.
- The system shall allow Admin to mark a menu item as available or unavailable.
- The system shall prevent unavailable menu items from being added to new orders.
- The system shall allow Waiter and Manager to view available menu items while creating an order.
- The system shall allow Admin to update menu item price.
- The system shall use the current menu item price while adding the item to an order.

- The system shall allow Admin to define basic item type such as vegetarian or non-vegetarian if required.

❏ Points To Remember:

- Menu price should be stored in the order item when the order is created.
Because if Admin changes the price tomorrow, old orders should not change.

Order Management

This module will allow Waiter, Manager, and authorized users to create and manage dine-in orders. The order management module will connect table selection, menu item selection, kitchen preparation, and billing.

- The system shall allow Waiter or Manager to create a dine-in order for an available table.
- The system shall allow Waiter or Manager to select a table while creating an order.
- The system shall allow Waiter or Manager to add menu items to an order.
- The system shall allow Waiter or Manager to update item quantity before sending the item to the kitchen.
- The system shall allow Waiter or Manager to add item-level notes or preparation instructions.
- The system shall allow Waiter or Manager to remove an item before it is sent to the kitchen.
- The system shall allow Waiter or Manager to send order items to the kitchen.
- The system shall assign an initial status to each order item when it is added to an order.
- The system shall allow Waiter or Manager to view active orders.
- The system shall allow Waiter or Manager to view order details including table, items, quantity, price, notes, and item status.
- The system shall prevent unavailable menu items from being added to an order.
- The system shall prevent creating more than one active dine-in order for the same table at the same time.
- The system shall allow additional items to be added to an active order.
- The system shall allow Manager to cancel an order or order item with a reason.

- The system shall require Manager approval for cancellation of items already sent to the kitchen.
- The system shall update the order total based on added, updated, or cancelled items.
- The system shall maintain the status of the order based on item progress and billing status.
- The system shall allow Waiter or Manager to request bill generation for an active order.

☐ **Order Status**

- Open
- In Progress
- Bill Requested
- Paid
- Cancelled

☐ **Item Status**

- Pending
- Preparing
- Ready
- Served
- Cancelled

☐ **Points To Remember**

- Order item cancellation after sending to kitchen should require Manager approval.

Kitchen Management

This module will allow Chef/Kitchen Staff to view orders sent to the kitchen and update the preparation status of order items. The kitchen management module helps improve coordination between waiters and kitchen staff.

- The system shall allow Chef/Kitchen Staff to view order items sent to the kitchen.
- The system shall display kitchen orders with table number, order number, item name, quantity, notes, and current status.

- The system shall allow Chef/Kitchen Staff to view item-level preparation instructions.
- The system shall allow Chef/Kitchen Staff to update item status from Pending to Preparing.
- The system shall allow Chef/Kitchen Staff to update item status from Preparing to Ready.
- The system shall prevent Chef/Kitchen Staff from generating bills or accepting payments.
- The system shall prevent Chef/Kitchen Staff from modifying item price or order billing details.
- The system shall allow Waiter or Manager to view updated kitchen item status.
- The system shall allow Waiter or Manager to mark a Ready item as Served.
- The system shall display pending and preparing items separately if required.
- The system shall allow kitchen staff to filter or sort kitchen orders by order time or status.
- The system shall update the order progress based on the status of its order items.

☐ Recommended responsibility:

- Chef: Pending → Preparing → Ready
- Waiter: Ready → Served
- Manager: Can cancel if required

Billing and Payment Management

This module will allow Cashier, Manager, and authorized users to generate bills, verify order items, apply charges or discounts, record payments, and close completed orders.

- The system shall allow Cashier or Manager to generate a bill for an active order.
- The system shall allow Cashier or Manager to view order items before generating the bill.
- The system shall calculate the bill amount based on ordered items, quantities, and stored item prices.
- The system shall apply configured taxes to the bill amount.
- The system shall apply configured service charges if enabled.
- The system shall allow authorized users to apply a discount if permitted.

- The system shall calculate the final payable amount after tax, service charge, and discount.
 - The system shall allow Cashier or Manager to record payment for a bill.
 - The system shall support payment modes such as Cash, Card, UPI, and Other.
 - The system shall allow Cashier or Manager to mark a bill as Paid after payment is recorded.
 - The system shall allow Cashier or Manager to print or download the customer receipt.
 - The system shall display restaurant details on the receipt based on restaurant setup.
 - The system shall prevent payment from being recorded for a cancelled order.
 - The system shall prevent duplicate payment for the same bill.
 - The system shall close the order after the bill is fully paid.
 - The system shall mark the related table as Available after the order is closed and payment is completed.
 - The system shall allow Admin or Manager to view bill and payment history.
- Payment gateway is not included in version 1.
- The system will only record payment manually.

Inventory Management

This module will allow Inventory Manager, Admin, and authorized users to manage restaurant ingredients, stock levels, suppliers, purchase entries, and wastage records. The inventory module will help the restaurant track available stock and identify low-stock items.

- The system shall allow Inventory Manager or Admin to create ingredients or raw materials.
- The system shall allow Inventory Manager or Admin to update ingredient details.
- The system shall allow Inventory Manager or Admin to deactivate ingredients that are no longer used.
- The system shall allow Inventory Manager or Admin to define unit of measurement for each ingredient.
- The system shall allow Inventory Manager or Admin to add stock purchase entries.
- The system shall allow Inventory Manager or Admin to update current stock quantity.

- The system shall allow Inventory Manager or Admin to manage supplier details.
- The system shall allow Inventory Manager or Admin to set reorder level for ingredients.
- The system shall identify ingredients whose stock quantity is below the reorder level.
- The system shall allow Inventory Manager or Admin to record stock wastage.
- The system shall allow Inventory Manager or Admin to view stock history.
- The system shall allow Inventory Manager or Admin to view low-stock items.
- The system shall allow Inventory Manager or Admin to view basic inventory reports.
- The system shall record the user and date/time for stock updates and wastage entries.

Reports

This module will allow Admin, Manager, and authorized users to view basic reports related to sales, orders, payments, menu items, and inventory. Reports will help restaurant management understand daily operations and make better decisions.

- The system shall allow Admin or Manager to view daily sales reports.
- The system shall allow Admin or Manager to view order summary reports.
- The system shall allow Admin or Manager to view payment summary reports.
- The system shall allow Admin or Manager to view item-wise sales reports.
- The system shall allow Admin or Manager to view cancelled orders or cancelled item reports.
- The system shall allow Admin, Manager, or Inventory Manager to view basic inventory reports.
- The system shall allow Admin, Manager, or Inventory Manager to view low-stock reports.
- The system shall allow reports to be filtered by date range.
- The system shall display total orders, total sales amount, total discount, total tax, and total payment amount where applicable.
- The system shall allow authorized users to view reports based on their role permissions.
- The system shall allow Admin or Manager to export basic reports if required.

Audit and Activity Tracking

This module will track important user actions performed in the system. Audit and activity tracking will help improve accountability, traceability, and operational control.

- The system shall record the user who creates a staff user, menu item, table, order, bill, payment, or inventory entry.
- The system shall record the user who updates important records such as menu items, table details, order items, bills, payments, and stock entries.
- The system shall record the date and time when important records are created or updated.
- The system shall record cancellation actions for orders or order items.
- The system shall store the cancellation reason when an order or order item is cancelled.
- The system shall record the user who changes kitchen item status.
- The system shall record the user who generates a bill.
- The system shall record the user who records payment.
- The system shall record the user who updates stock quantity or records wastage.
- The system shall allow Admin or Manager to view activity history for important operations if required.

Non-Functional Requirements

The system should satisfy the following non-functional requirements to ensure performance, security, usability, reliability, and maintainability.

Performance

The system should load commonly used pages such as dashboard, table list, menu list, and active orders within an acceptable response time.

Security

The system should protect user accounts and restrict access based on assigned roles and permissions.

Usability

The system should provide a simple and user-friendly interface so restaurant staff can use it easily during daily operations.

Reliability

The system should work consistently during restaurant operating hours and should handle normal daily order and billing operations without failure.

Scalability

The system should be designed so that future support for multiple restaurants, branches, and additional modules can be added without major redesign.

Maintainability

The system should be structured in a way that allows developers to maintain, update, and extend features easily.

Data Backup and Recovery

The system should support regular data backup and recovery to prevent data loss.

Business Rules

The system shall follow the business rules defined below to ensure proper restaurant operation and data consistency.

BR-001: A staff user must have an assigned role to access the system.

BR-002: A deactivated user shall not be allowed to log in to the system.

BR-003: A table cannot have more than one active dine-in order at the same time.

BR-004: Only available menu items can be added to new orders.

BR-005: Menu item price should be stored in the order item at the time the item is added to an order.

BR-006: Changing the price of a menu item should not affect already created orders.

BR-007: Order items can be removed freely only before they are sent to the kitchen.

BR-008: Cancelling an order item after it is sent to the kitchen should require Manager approval.

BR-009: Chef/Kitchen Staff can update item status only up to Ready status.

BR-010: Waiter or Manager can mark a Ready item as Served.

BR-011: A bill can be generated only for an active order.

BR-012: Payment cannot be recorded for a cancelled order.

BR-013: Duplicate payment should not be allowed for the same bill.

BR-014: An order should be closed only after the bill is fully paid.

BR-015: A table should become Available only after the related order is closed and payment is completed.

BR-016: Stock quantity should not be updated without recording the user and date/time of the update.

BR-017: Stock wastage should be recorded with quantity, reason, user, and date/time.

BR-018: Only authorized users should be allowed to apply discounts.

BR-019: Only authorized users should be allowed to cancel orders or order items.

BR-020: Important actions such as order creation, cancellation, billing, payment, and stock update should be tracked for audit purposes.

System Workflows

Dine-In Order Flow

1. Customer arrives at the restaurant.
2. Waiter or Manager checks available tables.
3. Waiter or Manager assigns an available table to the customer.

4. Waiter creates a dine-in order for the selected table.
5. The system marks the selected table as Occupied.
6. Waiter adds menu items to the order.
7. Waiter adds item notes or preparation instructions if required.
8. Waiter sends the order items to the kitchen.
9. The system updates order item status as Pending.
10. Kitchen staff views the order items.
11. Kitchen staff prepares the items and marks them as Ready.
12. Waiter serves the ready items and marks them as Served.
13. Waiter requests bill generation.
14. Cashier generates the bill.
15. Customer makes payment.
16. Cashier records the payment.
17. The system marks the order as Paid/Closed.
18. The system marks the table as Available.

Kitchen Preparation Flow

1. Waiter or Manager sends order items to the kitchen.
2. The system displays the sent order items on the kitchen screen.
3. Kitchen Staff views order details such as table number, order number, item name, quantity, and preparation notes.
4. The system shows the initial status of the order item as Pending.
5. Kitchen Staff starts preparing the item.
6. Kitchen Staff updates the item status from Pending to Preparing.
7. Waiter or Manager can view the updated item status.
8. After preparation is completed, Kitchen Staff updates the item status from Preparing to Ready.

9. Waiter or Manager can view the item as Ready.
10. Waiter serves the item to the customer.
11. Waiter updates the item status from Ready to Served.
12. The system updates the overall order progress based on the status of its order items.

Billing and Payment Flow

1. Customer requests the bill after completing the order.
2. Waiter or Manager requests bill generation for the active order.
3. Cashier opens the billable order.
4. Cashier verifies the ordered items, quantities, prices, and served items.
5. The system calculates the subtotal based on order items and stored item prices.
6. The system applies configured taxes if applicable.
7. The system applies configured service charges if enabled.
8. Authorized user applies discount if permitted.
9. The system calculates the final payable amount.
10. Cashier confirms and generates the bill.
11. Customer makes payment using Cash, Card, UPI, or other supported payment mode.
12. Cashier records the payment mode and payment amount.
13. The system marks the bill as Paid after successful payment recording.
14. The system closes the related order.
15. The system marks the related table as Available.
16. Cashier prints or downloads the customer receipt if required.
17. The system stores the bill and payment details for reports and history.

Menu Management Flow

1. Admin logs in to the system.
2. Admin opens the Menu Management module.
3. Admin creates or selects a menu category.
4. Admin adds a new menu item under the selected category.
5. Admin enters menu item details such as item name, description, price, item type, and availability.
6. Admin uploads an item image if required.
7. Admin saves the menu item.
8. The system validates the required menu item details.
9. The system stores the menu item information.
10. The menu item becomes available for order creation if it is marked as available.
11. Waiter or Manager can view the available menu item while creating an order.
12. Admin can update menu item details, price, image, or availability when required.
13. If a menu item is marked as unavailable, the system prevents it from being added to new orders.
14. Existing orders should not be affected by changes made to menu item price or availability.

Inventory Stock Update Flow

1. Inventory Manager or Admin logs in to the system.
2. Inventory Manager opens the Inventory Management module.
3. Inventory Manager selects an existing ingredient or creates a new ingredient if it does not already exist.
4. Inventory Manager enters ingredient details such as ingredient name, unit of measurement, current stock quantity, reorder level, and supplier details.
5. Inventory Manager adds a stock purchase entry when new stock is purchased.

6. The system updates the current stock quantity based on the purchase entry.
7. Inventory Manager can manually update stock quantity if correction is required.
8. If any stock is damaged, expired, or wasted, Inventory Manager records the wastage quantity and reason.
9. The system reduces the available stock quantity based on the wastage entry.
10. The system checks whether the current stock quantity is below the defined reorder level.
11. If stock is below the reorder level, the system marks the ingredient as low stock.
12. Inventory Manager or Admin can view stock history, purchase entries, wastage entries, and low-stock items.
13. The system records the user and date/time for stock updates, purchase entries, and wastage records.
14. Inventory reports are updated based on stock purchases, stock updates, and wastage entries.

In version 1, stock update will be handled manually through purchase entries, stock adjustments, and wastage entries.

Recipe-based automatic stock deduction will not be included in the first version.

Data Requirements

The system shall store and manage data related to users, roles, permissions, restaurant setup, tables, menu categories, menu items, orders, order items, bills, payments, ingredients, suppliers, stock transactions, and activity logs.

The main data entities of the system are:

User

Stores staff details such as name, email, mobile number, password hash, assigned role, status, created date, and updated date.

Role

Stores system roles such as Admin, Manager, Waiter, Chef, Cashier, and Inventory Manager.

Permission

Stores permissions that define what actions can be performed in the system.

RolePermission

Stores the mapping between roles and permissions.

Restaurant

Stores restaurant profile information such as name, address, contact details, logo, opening time, closing time, and default currency.

ChargeSetting

Stores tax and service charge settings.

RestaurantTable

Stores table number, table name, seating capacity, table status, and active status.

MenuCategory

Stores menu category details such as category name, description, display order, and active status.

MenuItem

Stores item details such as category, item name, description, price, image, item type, availability, and active status.

Order

Stores order details such as order number, table, order type, order status, subtotal, discount, tax, service charge, total amount, and created user.

OrderItem

Stores order item details such as menu item, item name, quantity, unit price, total price, item status, item note, cancellation status, and cancellation reason.

Bill

Stores bill details such as bill number, order, subtotal, discount, tax, service charge, final amount, bill status, generated user, and generated date.

Payment

Stores payment details such as bill, payment mode, payment amount, payment status, paid date, and received user.

Ingredient

Stores ingredient details such as ingredient name, unit of measurement, current stock quantity, reorder level, and active status.

Supplier

Stores supplier details such as supplier name, contact person, contact number, email, address, and active status.

StockTransaction

Stores stock purchase, adjustment, and wastage details such as ingredient, supplier, transaction type, quantity, cost, reason, transaction date, and created user.

ActivityLog

Stores important user actions such as order creation, item cancellation, bill generation, payment recording, stock update, and menu item update.

Main Relationships

- User belongs to one Role.
- Role can have many Permissions through RolePermission.
- MenuCategory can have many MenuItems.
- RestaurantTable can have many Orders over time.
- Order belongs to one Table.
- Order can have many OrderItems.
- OrderItem belongs to one MenuItem.
- Order can have one Bill.
- Bill can have one or more Payments.
- Ingredient can have many StockTransactions.
- Supplier can have many StockTransactions.

- User can create or update many records.

External Interface Requirements

This section defines the interfaces through which users, devices, or external systems interact with the Restaurant Management System.

User Interface

The system shall provide a web-based user interface for restaurant staff such as Admin, Manager, Waiter, Chef/Kitchen Staff, Cashier, and Inventory Manager.

The user interface should be simple, responsive, and easy to use during restaurant operations.

The system shall provide separate screens or views based on user roles and permissions.

The system shall provide screens for login, dashboard, table management, menu management, order management, kitchen view, billing, inventory, and reports.

Hardware Interface

The system may support bill printing through a connected printer.

The system should allow bills or receipts to be printed or downloaded.

Advanced hardware integrations such as barcode scanners, cash drawers, POS terminals, or kitchen display devices are not included in the first version.

Software Interface

The first version of the system will not include direct integration with third-party food delivery platforms, online payment gateways, accounting software, or customer-facing applications.

The system may internally use APIs between frontend and backend services for data communication.

Communication Interface

The system shall communicate between frontend and backend using secure HTTP/HTTPS APIs.

The system should use proper request and response formats for data exchange.

The system should require authenticated requests for protected operations.

Assumptions and Constraints

This section defines the assumptions and constraints considered for the first version of the Restaurant Management System.

Assumptions

1. The restaurant staff will have basic knowledge of using a web-based application.
2. The restaurant will configure basic data such as users, tables, menu categories, menu items, tax settings, and inventory items before daily operations.

3. The system will be used mainly by restaurant staff such as Admin, Manager, Waiter, Chef/Kitchen Staff, Cashier, and Inventory Manager.
4. The restaurant will have internet or local network connectivity during operating hours.
5. The Admin will be responsible for managing staff users, roles, menu data, table data, and restaurant settings.
6. Each staff user will be assigned a valid role before accessing the system.
7. Payments will be collected outside the system and recorded manually in the system.
8. Inventory stock updates will be entered manually by authorized users.
9. Menu item prices, taxes, and service charges will be configured correctly by authorized users.
10. Restaurant staff will update order, kitchen, billing, and inventory status accurately during daily operations.
11. Bill printing will depend on the availability and configuration of the connected printer or browser print option.
12. Reports will be generated based on the data entered and maintained in the system.

Constraints

1. The first version will support single restaurant operations only.
2. The first version will focus mainly on dine-in order management.
3. Customer-facing online ordering will not be included in the first version.
4. Delivery order management and live delivery tracking will not be included in the first version.
5. Online payment gateway integration will not be included in the first version.

6. Full SaaS onboarding and multi-branch management will not be included in the first version.
7. Advanced recipe-based automatic stock deduction will not be included in the first version.
8. Advanced accounting, payroll, and tax filing features will not be included in the first version.
9. The system will depend on user roles and permissions for access control.
10. The first version will not support advanced hardware integrations such as barcode scanners, cash drawers, POS terminals, or dedicated kitchen display systems.
11. The first version will not support offline mode or automatic data synchronization when the network is unavailable.
12. The accuracy of reports will depend on correct and timely data entry by restaurant staff.
13. The first version will provide basic reports only and will not include advanced analytics, forecasting, or AI-based recommendations.